QUESTIONS 1

(1) If $A = \{2, -3\}, B = \{x: x \text{ is the root of } x^2 - x - 2 = 0\}$. Show that A and B are not disjoint.

(2) Given the set $A = \{x, y, z\}$, Find the power set of A.

(3) Let $A = (0, 1]$ and $B = [4, \infty)$. With respect to the real numbers set, R, find (i) A' (ii) B' (iii) $A - B$ (iv) $B - A$ (v) $A \Delta B$ (vi) $A \cap B$ (vii) $A \cup B$.

(4) Represent (i) $A - B$ (ii) $A \Delta B$ on a Venn diagram.

(5) In a class of 800 students, 200 offered Chemistry and 300 offered Physics; 150 offered both Chemistry and Physics. Find how many students offered neither Chemistry nor Physics.

(6) Out of 158 television viewers interviewed on the NBS Television Station, 94 watch NTA and 85 watch the NBS station. How many watch both?

(7) In an examination, 36 boys offered French, 32 offered German and 31 offered Spanish, 9 offered French and German, 12 offered German and Spanish, and 11 offered French and Spanish. There were 4 boys who offered all three languages and 9 boys offered none of the languages. Let F, G and S represent the set of boys who offered French, German and Spanish respectively.

a. Represent the information on a Venn diagram and find how many boys are there altogether.

b. How many boys offered (i) at least one of the language (ii) at most two of the languages (iii) exactly one language (iv) exactly two languages?

c. Find $F' \cap G' \cap S'$ and interpret it.

d. Find the probability of boys who offered (i) French or German but not Spanish (ii) French and German but not Spanish.

e. What percentage of the boys offered (i) French only (ii) German only?

(8) Let U be the set of natural numbers, $\mathbb{N}$, $A = \{t: t = 2n + 1, n \in \{\mathbb{N} \cup \{0\}\}\}$ and $B = \{x: 0 < x < 10, x \in \mathbb{N}\}$. List the elements of (i) A (ii) A (iii) $A \cap B$ (iv) $A' \cap B'$.

QUESTIONS 2

(1) Graph the following inequalities on the number line.

(i) $\{x: -2 < x < 5\}$ (ii) $\{x: -2 < x < 5\}$ (iii) $\{x: 1 < x < 7\}$ (iv) $\{z: -4 < z < 3, z \in \mathbb{Z}\}$

(2) Prove the following by mathematical induction

(i) $1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$

(ii) $1^3 + 2^3 + 3^3 + \cdots + n^3 = \left[\frac{1}{2}n\,(n+1)\right]^2$

(iii) $\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$

(iv) $3 + 3^2 + 3^3 + \cdots + 3^n = \frac{3}{2}(3^n - 1)$

(3) Use mathematical induction to prove De Moivre's theorem,

$[r(\cos\theta + i\sin\theta)]^n = r^n(\cos n\theta + i\sin n\theta)$

(4) Prove that $a + ar + ar^2 + ar^3 + \cdots + ar^{n-1} = \frac{a(1-r^n)}{1-r}$ where n is any positive integer and r are numbers with $r \neq 1$.

(5) Prove that $\sum_{r=1}^{n} \frac{1}{r(r+1)(r+2)} = \frac{n(n+3)}{4(n+1)(n+2)}$

(6) Show that $\frac{1+x}{1-2x} = 1 + \frac{3}{2}x + \frac{15}{8}x^2 + \cdots$

(7) By putting $x = 0.08$ in $(1+x)^{\frac{1}{2}}$ and its expansion, find $\sqrt{3}$ correct to 4 significant figures.

(8) By putting $\frac{1}{10}$ for x in $(1-x)^{\frac{1}{2}}$ and its expansion, find $\sqrt{10}$ correct to 6 significant figures.

(9) Use binomial expansion to find the value of $\sqrt{1.01}$ to 5 d.p.

(10) Expand $\sqrt{\frac{1+2x}{1-2x}}$ as a series in ascending powers of x up to and including the term in x^2. By putting $x = \frac{1}{100}$ find an appropriation for $\sqrt{51}.$

(11) Expand $(2-x)^{10}$ in ascending powers of x up to the term in x^3 and use the series to find the value of $(1.98)^{10}$ to the nearest whole number.

(12) Show that ${}^nC_r = {}^nC_{n-r}$.

(13) Write down the constant term in the binomial expansion of $\left(2x + \frac{1}{3x}\right)^{10}$.

(20). Use the first 3 terms of $\sqrt{4+3x}$ to appropriate $\sqrt{4.03}$ to 5 decimal places.

QUESTIONS 3

1. The 7th term of an A.P. is twice the third term find the relation between the first term and the common difference and prove that the 8th term is 3 times the second term.

2. The first term of an A.P. is 5. If the 10^{th} term is double the 3^{rd} term find the 15^{th} term.

3. If $6, p, q, 48$ are in G.P. find p and q.

4. The 3^{rd} and 6^{th} terms of a G.P. are 6 and 162 respectively. Find the 1^{st} and 4^{th} terms.

5. Find the two numbers where the arithmetic means is 10 and geometric mean is 8.

6. The 3^{rd} term of an A.P. is 7 and the sum of the first ten terms of the A.P. is 120. Find the A.P.

7. The first three terms of an A.P. are $2x - 1, x + 3,$ and $3x - 2$. Find x and the sum of the first 8 terms.

8. If the sum of infinity of the series $x + x^3 + x^5 + x^7 + \cdots$ is $\frac{2}{3}$, find the values of x.

9. If the fifth term of a Geometric Progression is $\frac{2}{27}$, and the eight term is $\frac{2}{729}$, find the first term, the common ratio and sum to infinity of the G.P.

10. A geometric series has first term 1 and the common ratio r is positive. If the sum of the first five terms is twice the sum of the terms from the 6^{th} to the 15^{th} inclusive. Prove that $r^5 = \frac{1}{2}\left(\sqrt{3 - 1}\right)$.

QUESTIONS 4

(1) Using the quadratic formula, solve:

(i) $x^2 - 6x - 7 = 0$ (ii) $a^2 - 6a - 5 = 0$

(iii) $x^2 - 3x + 4 = 2x^2 + 44x - 3$ (iv) $\frac{2a}{a+1} + \frac{1}{a} = 1$

(2) Use the discriminate to determine the nature of the roots of

(i) $x^2 - 3x + 4 = 0$ (ii) $3y^2 - 22y + 5 = 0$

(iii) $2x^2 - 3x + 1 = 0$ (iv) $\frac{2}{2x+1} + \frac{3x}{x-2} = 0$

(3) Solve the following quadratic equations

(i) $3x^2 = 2(x + 4)$ (ii) $\frac{5}{3}s^2 + 3s + 1 = 0$

(iii) $\frac{5}{w^2} - \frac{10}{w} + 2 = 0$

(4) A manufacturer of tin cans wishes to construct a cylindrical can of height 20cm and of capacity 3000cm^3. Find the inner radius of the can.

(5) Given that α and β are the roots of $x^2 + x - 1 = 0$. Form a quadratic equation whose roots are:

(i) $\alpha + 1$ and $\beta + 1$

(ii) $\frac{1}{\alpha}$ and $\frac{1}{\beta}$

(iii) $\frac{\alpha+1}{\beta+1}$ and $\frac{\beta+1}{\alpha+1}$

(6) For what value(s) of p will the equation $4x^2 = 2px + 25$ have equal roots.

QUESTIONS 5

1. Prove the following trigonometric identities

(i) $\frac{2\sin\theta+\sin 2\theta}{1-\cos 2\theta} = \frac{\sin\theta}{1-cos\theta}$ (ii) $\frac{1+\sec\theta}{2\sec\theta} = \cos^2\left(\frac{\theta}{2}\right)$

(iii) $\sin 7\theta + \sin\theta - 2\sin 2\theta\cos 3\theta = 4\cos^2 3\theta\sin\theta$

(iv) $\frac{\sin\theta}{1-\frac{\cos\theta}{\sin\theta}} + \frac{\cos\theta}{1-\frac{\cos\theta}{\sin\theta}} = \sin\theta + \cos\theta$

2. Solve for θ in the equation $\cos 2\theta = 2 - 3\sin\theta$, $0^\circ < \theta < 360^\circ$.

3. Prove that $\sin 7\theta\cos 2\theta - \sin 5\theta\cos 4\theta = \cos 3\theta\sin 2\theta$, hence or otherwise find solutions for θ in the equation $\sin 7\theta\cos 2\theta = \cos 3\theta\sin 2\theta$, where $0^\circ \le \theta \le 180^\circ$.

4. Find the general solution of the following equation $\sec^2 x = 2\tan x + 4$.

5. Given that $\cos x = \frac{1}{5}$, find all the possible values for $\frac{\sec x - \tan x}{\sin x}$

6. By writing $\sqrt{3}\cos\theta + \sin\theta$ in the form $r\cos(\theta-\alpha)$. Find r and α, hence solve the equation

$\sqrt{3\cos\theta} + \sin\theta = 1$ for $0 \le \theta \le 2\pi$.

7. Show that in surd form $\cos 22.5^\circ = \frac{1}{\sqrt{4-2\sqrt{2}}}$ and $\sin 22.5^\circ = \frac{\sqrt{2}-1}{\sqrt{4-2\sqrt{2}}}$

8. solve the equation $z^3 = 8$ where z is complex.